Drivers and Post-Event Analysis of the July 4, 2025, Flash Flood in Central Texas

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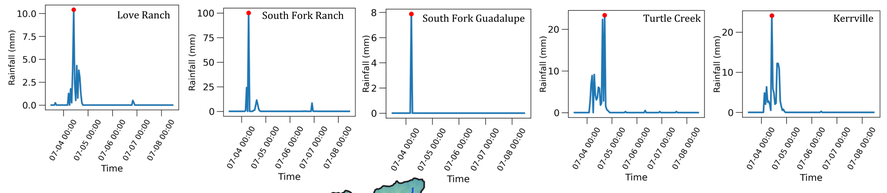
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Introduction

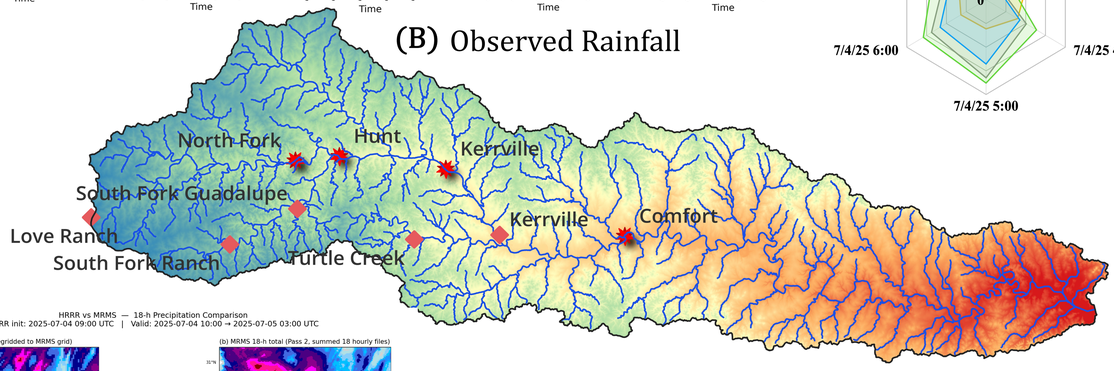
- On July 4th at dawn, a catastrophic flash flood hit Kerr and Kendall Counties in Texas. The event was triggered by intense rainfall up to 76-100 mm per hour in some of the impacted area, causing flash flooding with exceptionally high peaks throughout the Guadalupe River basin.
- We conducted this study to look at the causal mechanism of the event by utilizing multiple operational forecast products-HRRR rainfall forecast, NWM short-range streamflow forecast and Operational Flood Inundation Maps (FIMs).
- Finally, an impact-based and categorical assessment of the event is done across different lead times against the high-water mark-derived benchmark FIM.

Methodology

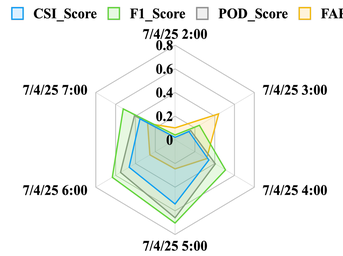
- We looked at the HRRR rainfall forecast at different lead hours and compared with the MRMS rainfall
- We employed the National Weather Service (NWS)-National Water Model (NWM) short-range forecast and the FIMserv tool (Baruah et al., 2025) to generate 306 FIMs at the Hydrologic Unit Code (HUC)-8 scale.
- For evaluation , we use post-event USGS observed high-water marks to create a spatially continuous FIM benchmark.

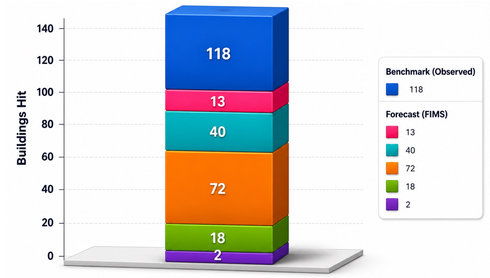


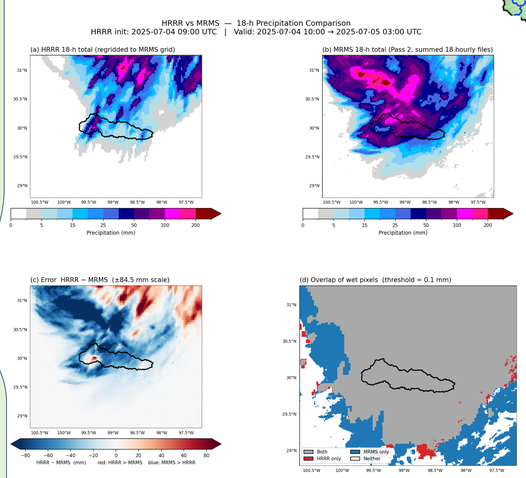
(B) Observed Rainfall



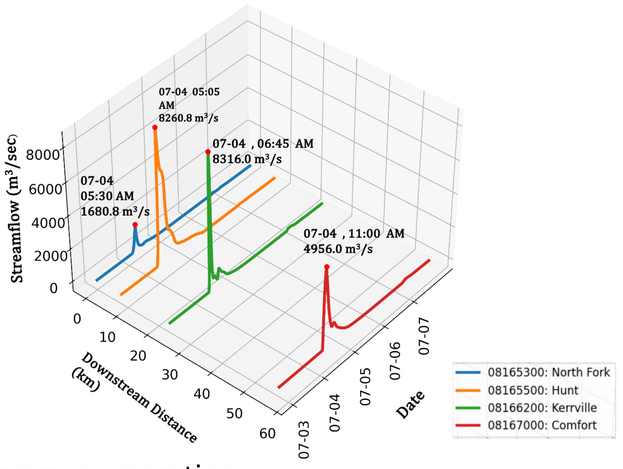
(A) Study Area



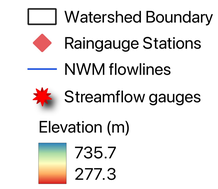


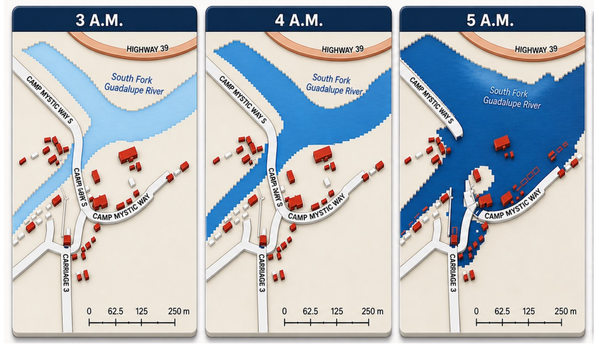


(C) HRRR forecast and MRMS rainfall estimates

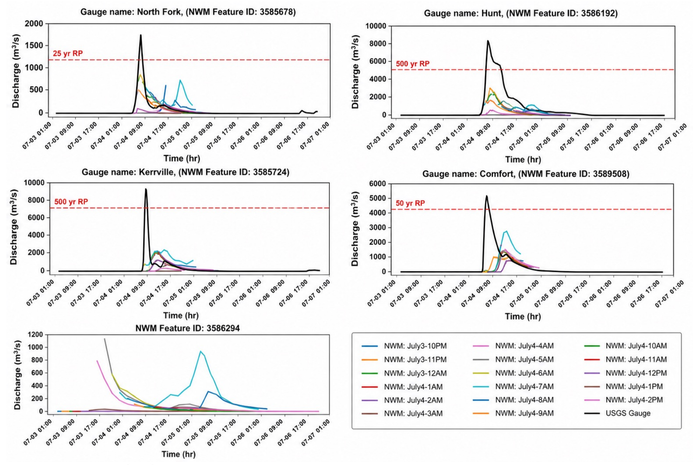


(D) Flood wave propagation





(F) FIM Evaluation and Impact Assessment



(E) NWM SRF and Gauge observation

Findings

- Rainfall intensities surpassed 100-year return period thresholds for durations ranging from 3 to 24 hours at the rainfall gauges when compared against NOAA Atlas 14 precipitation frequency curves (Perica et al., 2018; NSSL, 2025). HRRR persistently underestimated and spatially misplaced the convective cells relative to MRMS, and that displacement propagated cleanly through the hydrologic chain. The FIM error is largely a geometric error inherited from the QPF displacement.
- USGS gauge near Hunt on the Guadalupe River failed during the fast-rising streamflow. This outage disrupted the NWM's data assimilation process from performing streamflow "nudging" to correct model states toward observed discharge. These errors were extrapolated with downstream flow propagation, resulting in systematic underprediction of streamflow.
- Near Camp Mystic, forecast FIM skill scores (CSI, F1 score, and POD) improved as the forecast time approached the reference peak. It is also clear from the impact assessment that forecast FIM is not able to capture the majority of flooded buildings compared to the benchmark.